RAD Sem01_2627 · PART 1 PROJECT PROPOSAL

SaigonBite: one platform to spread lunch-hour orders across hubs

Online Food Delivery Hub Platform (OFDH)

Nguyen Thanh Dat · 722I0004
Owns UC-01 Place an order and UC-03 Manage hubs

Le Minh Quan · 523K0049
Owns UC-02 Deliver an order and UC-04 Reports

Team02 · Presented live on 26 October 2026

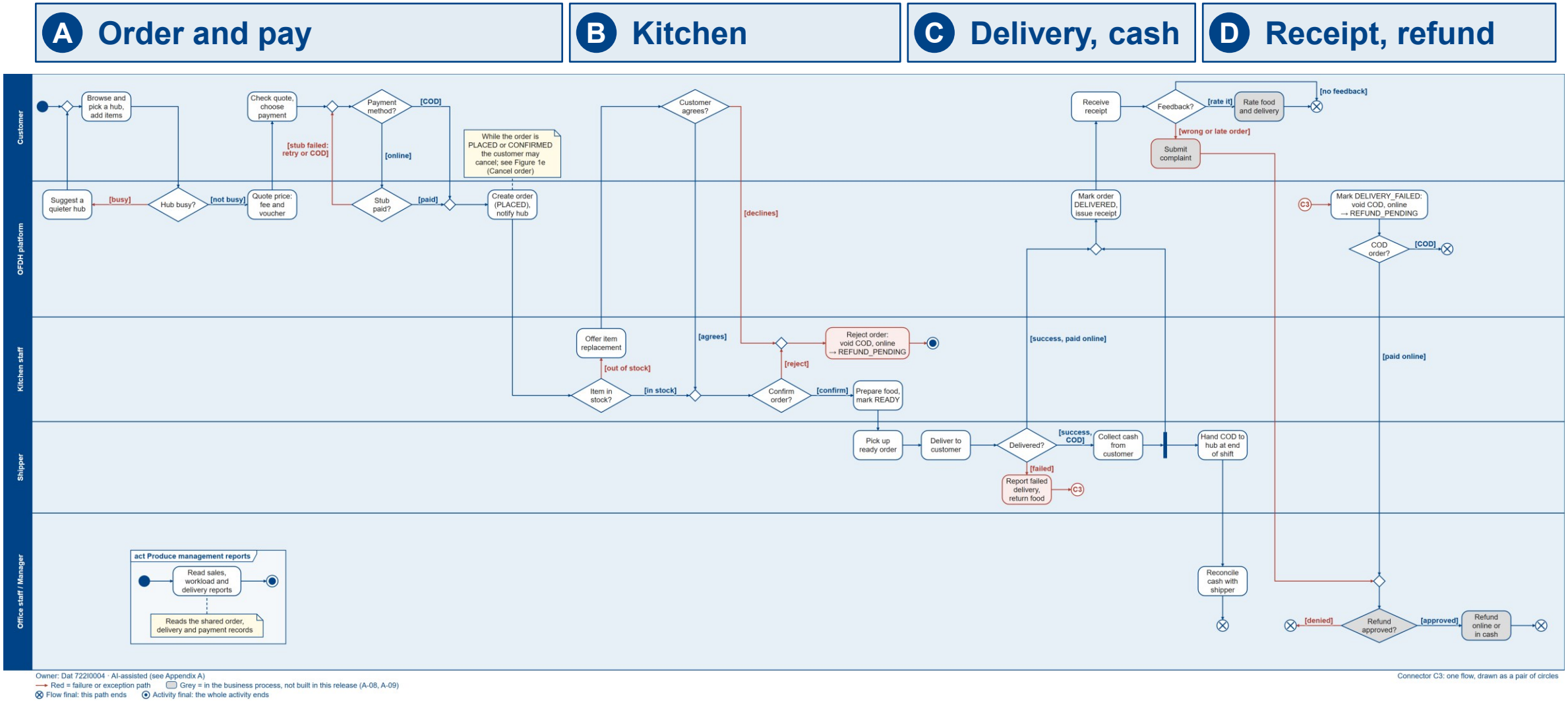
Paper tickets hide the queue: one hub overloads, another idles

Five pains from the case map to five capabilities of one shared platform.

Pain (from the case)	Capability in this release
One hub overloaded, another idle	Busy-hub block, quieter-hub suggestion
Cash handed over late, with mistakes	Cash recorded per order, reconciled
Ingredient runs out; staff phone customers	Reject or replace with consent
Tourists cannot register or pay	Local and tourist types; COD for all
No reliable reports	Reports from shared records

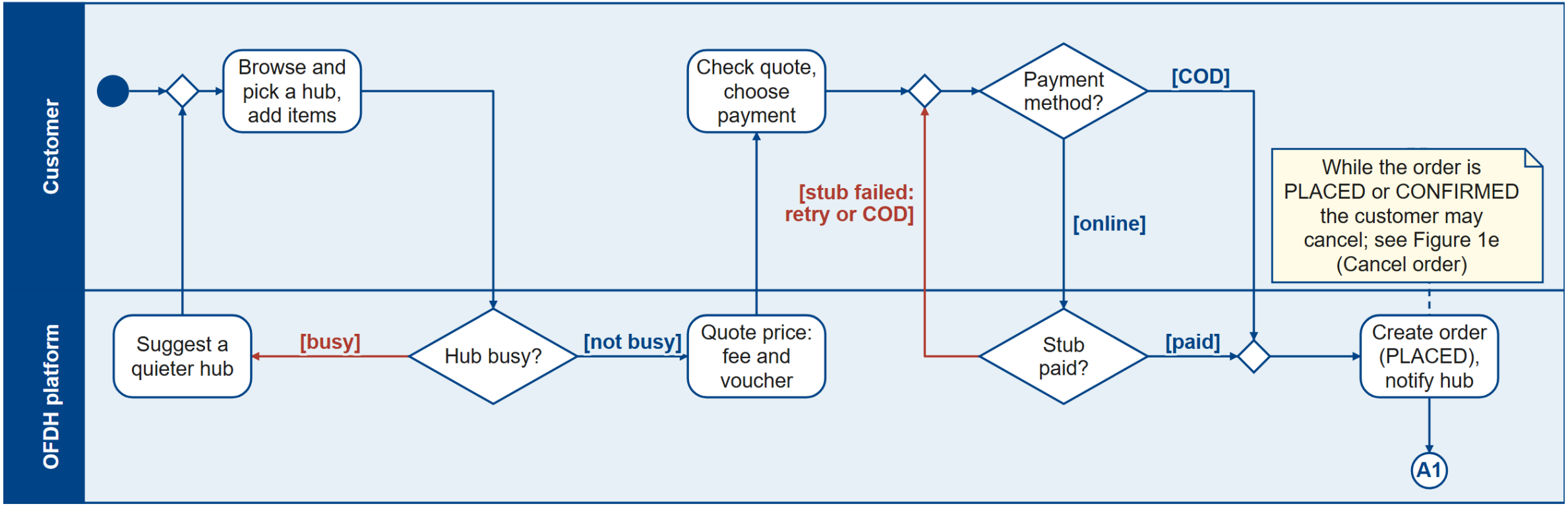
Vision One order record, visible workload, reports from the same data. Benefits are expected, not measured.

One order crosses five roles, and every branch has an owner



A busy hub redirects the customer before any order exists

Part A: Order, quote and payment



Owner: Dat 722I0004 · AI-assisted (see Appendix A)

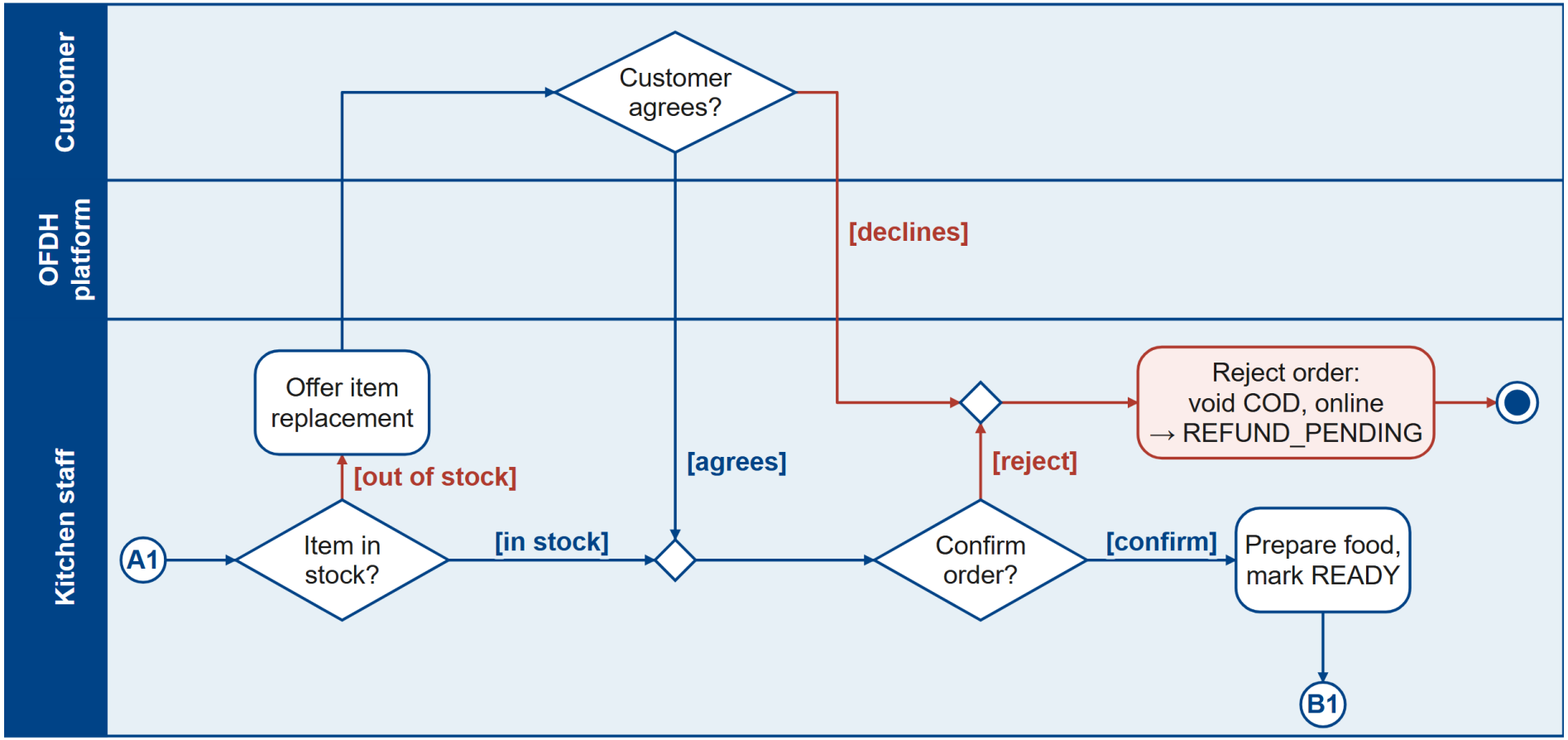
Connectors: A1 continues in Part B

→ Red = failure or exception path Grey = in the business process, not built in this release (A-08, A-09)

⊗ Flow final: this path ends ◉ Activity final: the whole activity ends

The kitchen gates the order; cancelling is its own short activity

Part B: Kitchen and item replacement



Owner: Dat 722I0004 · AI-assisted (see Appendix A) Connectors: A1 comes from Part A; B1 continues in Part C

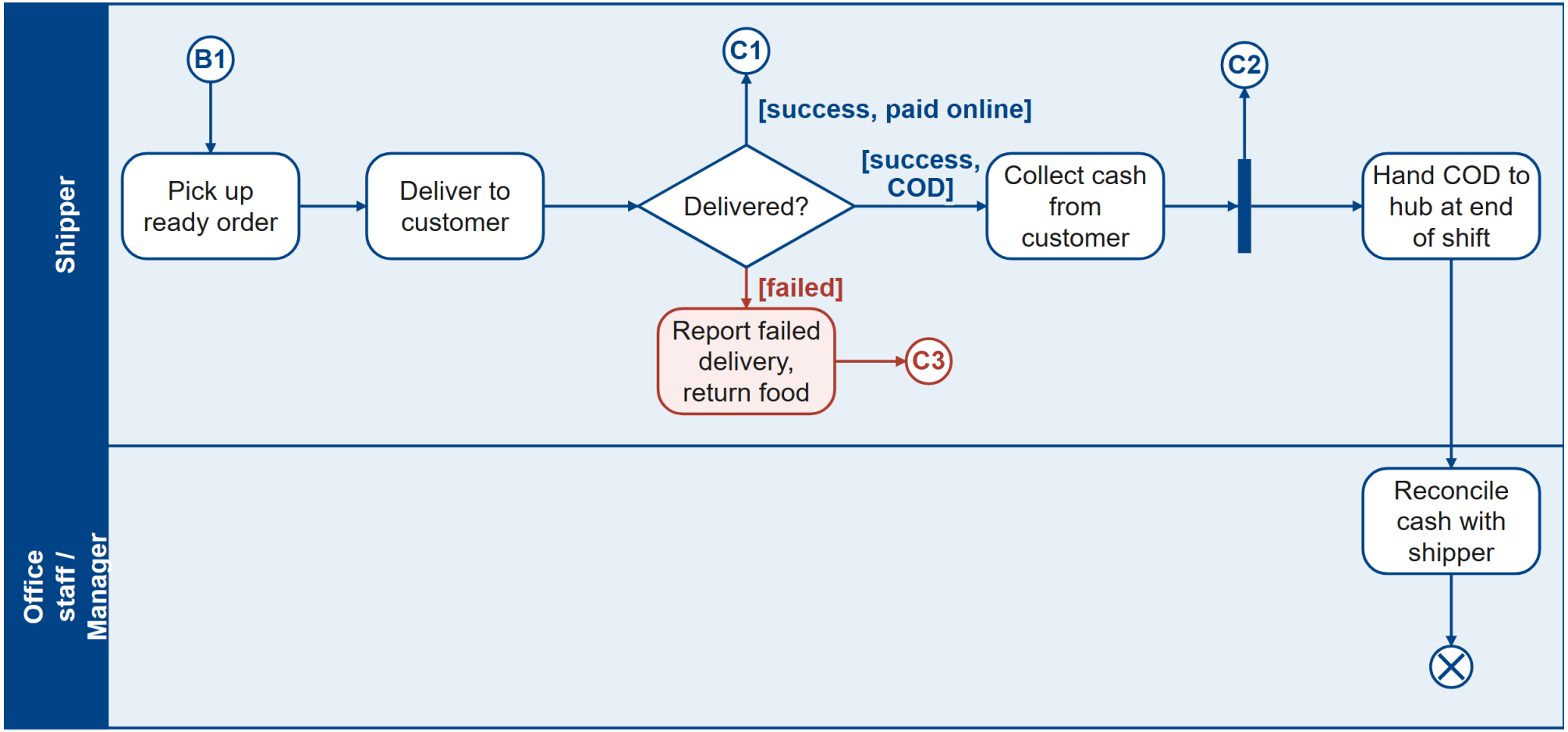
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Part E Cancel order is a separate activity (proposal Figure 1e): analysed only, A-08.

Delivered is not paid: cash is collected, then reconciled

Part C: Delivery and cash



Owner: Dat 722I0004 · AI-assisted (see Appendix A)

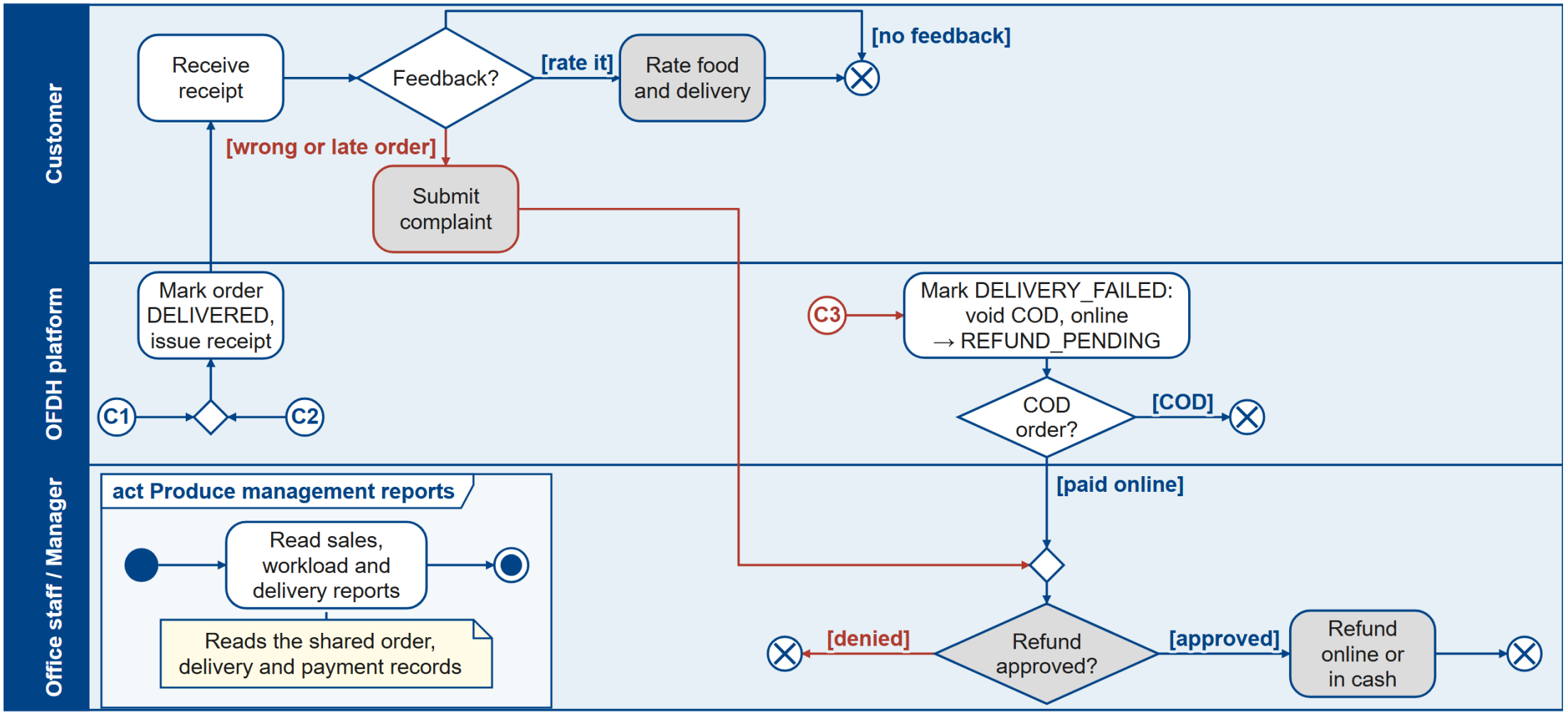
→ Red = failure or exception path Grey = in the business process, not built in this release (A-08, A-09)

⊗ Flow final: this path ends ◉ Activity final: the whole activity ends

Connectors: B1 comes from Part B; C1, C2, C3 continue in Part D

Only complaints and paid-online failures reach the refund decision

Part D: Receipt, failed delivery, refund and reports



Owner: Dat 722I0004 · AI-assisted (see Appendix A)

Connectors: C1, C2, C3 come from Part C

→ Red = failure or exception path Grey = in the business process, not built in this release (A-08, A-09)

⊗ Flow final: this path ends ◉ Activity final: the whole activity ends

Each actor decides one thing the order cannot move without

The shipper records COD collection and hands the cash to the hub; office staff reconcile it at the end of the shift.

Actor	Decides or does	Without them
Customer	Picks hub, items, payment	No order exists
Kitchen staff	Confirms or rejects, prepares	Nothing to deliver
Shipper	Delivers, records COD collection	Cash and order stay open
Admin	Sets hubs and capacity	Busy rule has no limit
Manager	Reads reports	Hubs cannot be compared
Office staff	Reconciles cash, decides refunds	Cash and complaints unchecked
Payment stub	Returns the online result	Online path untestable

Four use cases cover all four required groups, two per student

Each student owns two use cases and their state machines; five scope levels below.

ID	Use case	Actor	Group	Owner	State machine
UC-01	Place an order (#3)	Customer	(a) ordering	Dat	Order
UC-02	Deliver an order (#10)	Shipper	(b) fulfilment	Quan	Delivery
UC-03	Manage hubs, capacity (#13)	Admin	(c) full CRUD	Dat	Hub
UC-04	Produce reports (#16)	Manager	(d) report	Quan	none (read-only)

1 Built UC-01 to UC-04 in one application

2 Minimum support Kitchen board; login, seeded users

3 Analysed only Supporting use cases incl. cancel (A-08, L1)

4 Deferred Complaint, refund, rating (A-09, L3)

5 Future Loyalty, subscriptions, group orders

Iterative Unified Process fits a case whose rules are still open

Six time-boxed iterations: analysis, design, build, test and a joint review.

Why iterative

1 Answers pending
Client figures arrive later

2 One shared Order
Four use cases grow one record

3 Fixed deadlines
26/10, 16/11, 30/11; team of two

4 Riskiest first
Capacity in I2, payment in I3

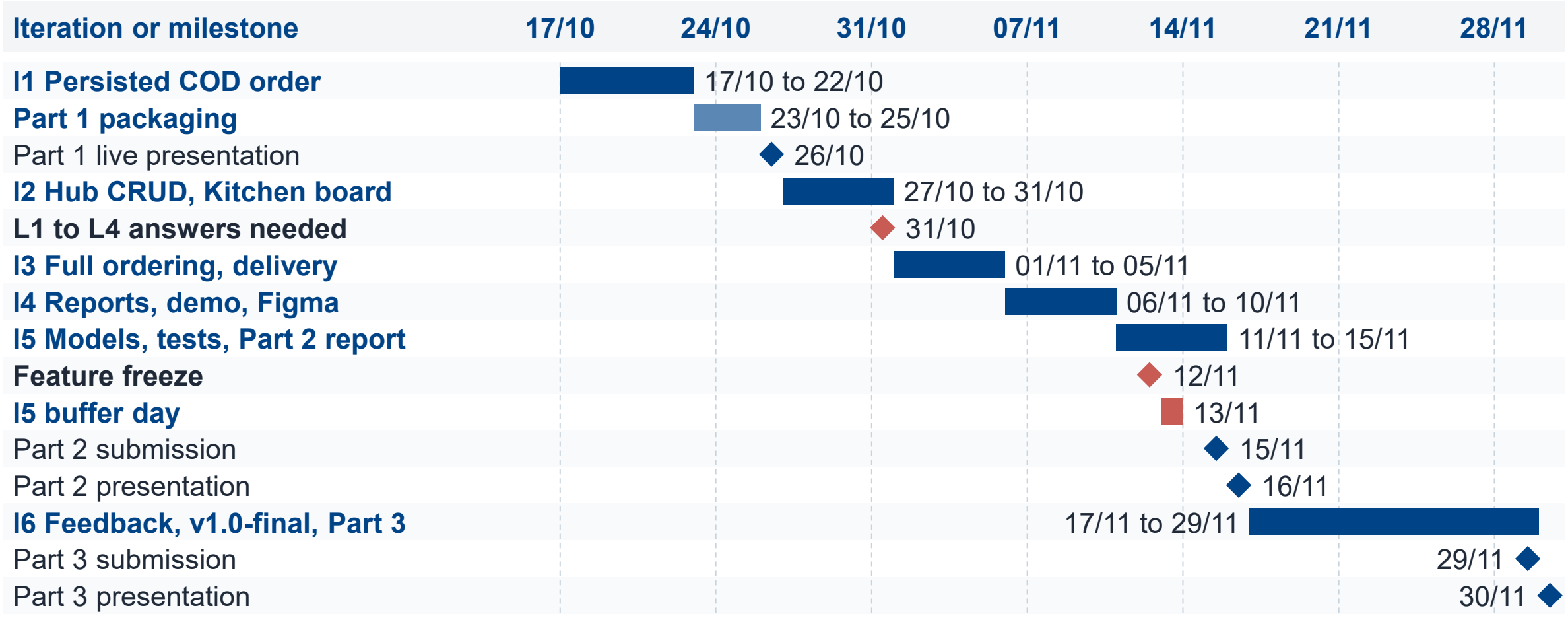
Stack and three layers (decided)

View	Flask + Jinja2 templates
Controller	One per use case
Domain	Order, Hub, Payment, Delivery
DAO	sqlite3, hand-written SQL
Database	SQLite, one database

Python 3.12 or later (I1 tested on 3.13)
pytest · Flask-Login

Six iterations land Part 2 on 15/11, with a freeze and a buffer day

Real dates 17/10 to 30/11; Part 2 does not rely on I6.



Contingency answers 31/10 · freeze 12/11 · buffer 13/11 · I6 is not a rescue buffer

Iteration 1 runs the six core processes in six days

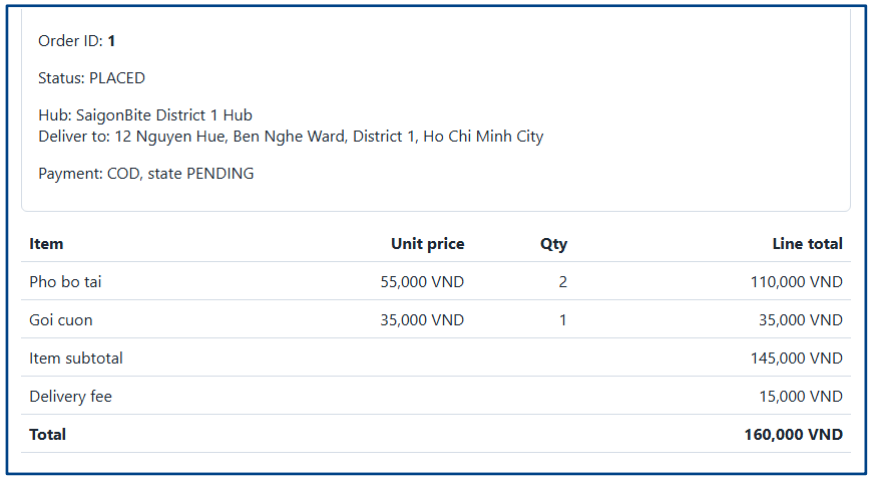
About 4 focused hours per student per day, 48 person-hours; setup is part of D1.

Day	Core process	Dat (UC-01)	Quan (foundation)	Evidence
D1 Sat 17/10	**CP1–2 Identify, plan**	Vision, backlog, UC-01 stories	Repo, skeleton, events, actors	Skeleton runs; backlog v1
D2 Sun 18/10	**CP3 Discover details**	UC-01 description, process diagram	Domain model, CRUD inventory	Names fixed
D3 Mon 19/10	**CP3–4 Design components**	Activity, SSD, Order states	Schema, seed, UC-02 draft	SSD and schema agree
D4 Tue 20/10	**CP4–5 Build, test**	Controller, Order code	DAOs, OrderDAO insert	COD order saved
D5 Wed 21/10	**CP5 Integrate**	Checkout page, pytest	Role checks, README	pytest run recorded
D6 Thu 22/10	**CP6 Test, deploy**	Fix defects, acceptance run	Second-machine run	Result; I1 review

Iteration 1: COD order persisted and tested; 7 of 9 criteria met

Real figures from the repository, the pytest output and the hours log.

Measure	Plan	Actual
Days with evidence	6	D3–D5 scope built on D3; D6 open
Person-hours	48	⟦ACTUAL s13_hours: hours⟧
Criteria met	9 of 9	7 of 9 (AC-1, AC-9 partial)
pytest	AC-4 to AC-8	81 passed, 0 failed (19/10)
Second machine	Passes	⟦ACTUAL s13_second_machine: result⟧
Roadmap change	None	⟦ACTUAL s13_roadmap_change: change⟧



Order ID: **1**

Status: PLACED

Hub: SaigonBite District 1 Hub
Deliver to: 12 Nguyen Hue, Ben Nghe Ward, District 1, Ho Chi Minh City

Payment: COD, state PENDING

Item	Unit price	Qty	Line total
Pho bo tai	55,000 VND	2	110,000 VND
Goi cuon	35,000 VND	1	35,000 VND
Item subtotal			145,000 VND
Delivery fee			15,000 VND
Total			**160,000 VND**

Saved order: ID 1, PLACED, COD, total 160,000 VND

Status Built early on 19/10 with AI assistance; team review pending.

Lesson ⟦ACTUAL lesson_learned: lesson from Iteration 1⟧

Unconfirmed rules and the capacity race are the biggest risks

Each risk has an owner and a concrete mitigation; none relies on I6.

ID	Risk	L / I	Owner	Mitigation
R-01	**CRUD coverage rejected**	Med / High	Dat	Answer by 31/10; generic admin fallback
R-02	**Rules unconfirmed or changed**	High / Med	Dat	Assumption Log; values in configuration
R-03	**Peak-hour capacity race**	Med / High	Dat	One BEGIN IMMEDIATE transaction; boundary test
R-04	**Integration drift between use cases**	Med / High	Dat	Shared schema, seed; demo at I3, I4
R-05	**Member unavailable or late**	Med / High	Quan	Cross-review; I5 buffer on 13/11

Contingency answers 31/10 · freeze 12/11 · buffer 13/11 · I6 is not a rescue buffer

Red marks the two risks in the title: R-02 and R-03.

Five client and four lecturer questions decide the open rules

Each question asks for one decision and carries our default.

Client questions (with our default)

Q1 Workload Busy count? Default: at capacity.

Q2 Delivery Distance, fee? Default: base + per km.

Q3 Cancel Until when? Default: before PREPARING.

Q4 Payment Online options? Default: wallet, card.

Q5 Reports Which date? Default: delivered date.

Questions for the lecturer

L1 Analysis plus seed data enough?

L2 Team-built payment stub acceptable?

L3 Defer complaint, refund, rating?

L4 Python 3.12 or later acceptable?

Close Answers needed by 31/10. A-10 confirmed verbally by the lecturer. Next: Iteration 2 on 27/10.